

MR-Hevo: theoretical derivations

LD adjustment for summary statistics; GATE score coefficient

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Part I

Recovering multivariate regression coefficients from univariate summary statistics via the LD correlation matrix

1 Introduction

In genome-wide association studies (GWAS), summary statistics are typically reported as *marginal* (univariate) regression coefficients $\hat{\beta}_{u,j}$, one per SNP, estimated by regressing the trait on each SNP separately. When a locus contains P SNPs in linkage disequilibrium (LD), the marginal coefficient for SNP j conflates its direct effect on the trait with indirect contributions absorbed from correlated neighbours.

This document derives the formula for recovering *multivariate* (joint) regression coefficients $\hat{\beta}_m$ from the marginal coefficients, using the LD correlation matrix among SNPs. The derivation works with **unstandardized** genotype units (allele dosage 0/1/2) throughout: both the input ($\hat{\beta}_u$) and the output ($\hat{\beta}_m$) are in units of trait change per additional copy of the effect allele. Allele frequencies are assumed available so that the genotype variances can be computed.

2 Notation

Let n be the number of individuals and P the number of SNPs in a locus.

- $\mathbf{g}_j \in \mathbb{R}^n$: vector of allele dosages for SNP j (centred, but **not** scaled to unit variance). Each $g_{ij} \in \{0, 1, 2\}$ minus the mean $2f_j$.
- $\sigma_j^2 = \text{Var}(g_j) = 2f_j(1 - f_j)$: genotype variance for SNP j with allele frequency f_j .
- $\mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_P)$: diagonal matrix of genotype standard deviations.
- $\mathbf{G} = [\mathbf{g}_1 \mid \dots \mid \mathbf{g}_P]$: $n \times P$ unstandardized genotype matrix.
- $\mathbf{G}^* = \mathbf{G}\mathbf{D}^{-1}$: genotype matrix with each column scaled to unit variance ($\mathbf{G}^* = [\mathbf{g}_1/\sigma_1 \mid \dots \mid \mathbf{g}_P/\sigma_P]$).
- $\mathbf{C}_g = \mathbf{G}^{*\top} \mathbf{G}^*/n$: $P \times P$ LD **correlation** matrix with $[\mathbf{C}_g]_{jj} = 1$ and $[\mathbf{C}_g]_{jk} = r_{jk}$ (the Pearson correlation between SNPs j and k).
- $\Sigma_g = \mathbf{G}^\top \mathbf{G}/n$: $P \times P$ **covariance** matrix of the unstandardized genotypes.
- $\mathbf{y} \in \mathbb{R}^n$: centred trait vector.

The covariance and correlation matrices are related by

$$\Sigma_g = \mathbf{D} \mathbf{C}_g \mathbf{D}. \quad (1)$$

3 Multivariate regression: the target

The joint ordinary-least-squares (OLS) estimator $\hat{\beta}_m \in \mathbb{R}^P$ in

$$\mathbf{y} = \mathbf{G}\beta_m + \varepsilon$$

is given by the normal equations

$$(\mathbf{G}^\top \mathbf{G}) \hat{\boldsymbol{\beta}}_m = \mathbf{G}^\top \mathbf{y}. \quad (2)$$

Dividing both sides by n :

$$\boldsymbol{\Sigma}_g \hat{\boldsymbol{\beta}}_m = \frac{1}{n} \mathbf{G}^\top \mathbf{y}, \quad (3)$$

so $\hat{\boldsymbol{\beta}}_m = \boldsymbol{\Sigma}_g^{-1}(\mathbf{G}^\top \mathbf{y}/n)$ whenever $\boldsymbol{\Sigma}_g$ is invertible. The entries $\hat{\beta}_{m,j}$ are in units of **trait change per allele** — the same units as the marginal coefficients.

4 Marginal regression coefficients

The univariate OLS estimate from regressing $\mathbf{y}$ on $\mathbf{g}_j$ alone is

$$\hat{\beta}_{u,j} = \frac{\mathbf{g}_j^\top \mathbf{y}}{\mathbf{g}_j^\top \mathbf{g}_j} = \frac{\mathbf{g}_j^\top \mathbf{y}}{n \sigma_j^2}. \quad (4)$$

Rearranging, the sample covariance of $\mathbf{g}_j$ with $\mathbf{y}$ is

$$\frac{\mathbf{g}_j^\top \mathbf{y}}{n} = \sigma_j^2 \hat{\beta}_{u,j}. \quad (5)$$

Stacking over all j , with $\mathbf{D}^2 = \text{diag}(\sigma_1^2, \dots, \sigma_P^2)$:

$$\frac{1}{n} \mathbf{G}^\top \mathbf{y} = \mathbf{D}^2 \hat{\boldsymbol{\beta}}_u. \quad (6)$$

5 Key identity: multivariate from marginal via the inverse correlation matrix

Substituting (6) into (3) and using the factorization (1):

$$\boldsymbol{\Sigma}_g \hat{\boldsymbol{\beta}}_m = \mathbf{D}^2 \hat{\boldsymbol{\beta}}_u \implies \mathbf{D} \mathbf{C}_g \mathbf{D} \hat{\boldsymbol{\beta}}_m = \mathbf{D}^2 \hat{\boldsymbol{\beta}}_u.$$

Premultiplying both sides by $\mathbf{D}^{-1}$:

$$\mathbf{C}_g \mathbf{D} \hat{\boldsymbol{\beta}}_m = \mathbf{D} \hat{\boldsymbol{\beta}}_u.$$

Premultiplying both sides by $\mathbf{C}_g^{-1}$:

$$\mathbf{D} \hat{\boldsymbol{\beta}}_m = \mathbf{C}_g^{-1} \mathbf{D} \hat{\boldsymbol{\beta}}_u.$$

Finally, premultiplying by $\mathbf{D}^{-1}$:

$$\boxed{\hat{\boldsymbol{\beta}}_m = \mathbf{D}^{-1} \mathbf{C}_g^{-1} \mathbf{D} \hat{\boldsymbol{\beta}}_u.} \quad (7)$$

Both $\hat{\boldsymbol{\beta}}_u$ and $\hat{\boldsymbol{\beta}}_m$ are in **units of trait change per allele**; $\mathbf{D}$ converts between per-allele and per-SD units internally.

5.1 Interpretation via per-SD coefficients

It is instructive to introduce per-SD (standardized) marginal coefficients $\hat{\beta}_{u,j}^* = \sigma_j \hat{\beta}_{u,j}$, i.e.

$\hat{\boldsymbol{\beta}}_u^* = \mathbf{D} \hat{\boldsymbol{\beta}}_u$, and per-SD multivariate coefficients $\hat{\boldsymbol{\beta}}_m^* = \mathbf{D} \hat{\boldsymbol{\beta}}_m$. Then equation (7) decomposes into three steps:

1. **Scale** marginal coefficients to per-SD units: $\hat{\boldsymbol{\beta}}_u^* = \mathbf{D} \hat{\boldsymbol{\beta}}_u$.
2. **Apply** the inverse correlation matrix: $\hat{\boldsymbol{\beta}}_m^* = \mathbf{C}_g^{-1} \hat{\boldsymbol{\beta}}_u^*$.
3. **Rescale** back to per-allele units: $\hat{\boldsymbol{\beta}}_m = \mathbf{D}^{-1} \hat{\boldsymbol{\beta}}_m^*$.

Steps 1–2 recover the per-SD multivariate coefficients (the formula for the fully standardized case); step 3 converts back to the original per-allele units.

5.2 Verification

We can verify equation (7) by checking that $\hat{\beta}_m$ satisfies the normal equations (3):

$$\Sigma_g \hat{\beta}_m = \mathbf{D} \mathbf{C}_g \mathbf{D} \cdot \mathbf{D}^{-1} \mathbf{C}_g^{-1} \mathbf{D} \hat{\beta}_u = \mathbf{D} \mathbf{C}_g (\mathbf{C}_g^{-1}) \mathbf{D} \hat{\beta}_u = \mathbf{D}^2 \hat{\beta}_u = \frac{1}{n} \mathbf{G}^\top \mathbf{y}. \quad \checkmark$$

6 Application to GWAS summary statistics

In practice the discovery GWAS provides only per-allele marginal coefficients $\hat{\beta}_{u,j}$ and their standard errors; individual genotypes are unavailable. The LD correlation matrix $\mathbf{C}_g$ is estimated from a reference panel of n_{ref} individuals with similar ancestry (e.g. 1000 Genomes EUR, $n_{\text{ref}} = 503$):

$$\hat{\mathbf{C}}_g = \hat{\mathbf{G}}^{*\top} \hat{\mathbf{G}}^* / n_{\text{ref}},$$

where $\hat{\mathbf{G}}^*$ is the reference panel genotype matrix with each column standardized to unit variance. The per-SNP SDs $\hat{\sigma}_j$ are likewise taken from the reference panel. Equation (7) then gives

$$\hat{\beta}_m = \hat{\mathbf{D}}^{-1} \hat{\mathbf{C}}_g^{-1} \hat{\mathbf{D}} \hat{\beta}_u, \quad \hat{\mathbf{D}} = \text{diag}(\hat{\sigma}_1, \dots, \hat{\sigma}_P).$$

This approximation is exact when the reference-panel LD matches the discovery-sample LD.

6.1 Scalar instrument construction

For Mendelian randomization, the P per-allele multivariate coefficients at a locus are combined into a single scalar instrument. Define the locus score for individual i :

$$S_i = \mathbf{g}_i^\top \hat{\beta}_m,$$

and the scalar instrument-exposure coefficient

$$\hat{\alpha} = \|\hat{\beta}_m\|,$$

i.e. the Euclidean norm of the **per-allele** multivariate coefficient vector. The scalar instrument is then

$$Z_i = \frac{S_i}{\hat{\alpha}} = \frac{\mathbf{g}_i^\top \hat{\beta}_m}{\|\hat{\beta}_m\|}.$$

Justification. In expectation, $X = \mathbf{g}^\top \beta_m + \varepsilon_X$, so

$$\text{Cov}(X, Z) = \frac{\text{Var}(\mathbf{g}^\top \beta_m)}{\hat{\alpha}} = \frac{\beta_m^\top \Sigma_g \beta_m}{\hat{\alpha}}, \quad \text{Var}(Z) = \frac{\beta_m^\top \Sigma_g \beta_m}{\hat{\alpha}^2}.$$

The regression coefficient of X on Z (through the origin) is therefore

$$\frac{\text{Cov}(X, Z)}{\text{Var}(Z)} = \frac{\beta_m^\top \Sigma_g \beta_m / \hat{\alpha}}{\beta_m^\top \Sigma_g \beta_m / \hat{\alpha}^2} = \hat{\alpha},$$

confirming $\mathbb{E}[X \mid Z] \approx \hat{\alpha} Z$.

6.2 Standard error of $\hat{\alpha}$

The Fisher information about α from N observations of the instrument-exposure regression $X = \alpha Z + \varepsilon_X$ is

$$\mathcal{I}_\alpha = N \frac{\text{Var}(Z)}{\text{Var}(\varepsilon_X)},$$

where $\text{Var}(\varepsilon_X) = \sigma_X^2 - \alpha^2 \text{Var}(Z)$ is the residual variance of X after projecting onto Z .

Evaluating $\text{Var}(Z)$. With $\hat{\alpha} = \|\hat{\beta}_m\|$:

$$\text{Var}(Z) = \frac{\hat{\beta}_m^\top \Sigma_g \hat{\beta}_m}{\|\hat{\beta}_m\|^2},$$

estimated empirically from the reference panel as $\widehat{\text{Var}}(Z) = \widehat{\text{Var}}(S) / \hat{\alpha}^2$, where $\widehat{\text{Var}}(S) = n_{\text{ref}}^{-1} \sum_i S_i^2$ and $S_i = \mathbf{g}_i^\top \hat{\beta}_m$ are the reference-panel scores.

Note that

$$\hat{\alpha}^2 \text{Var}(Z) = \hat{\beta}_m^\top \Sigma_g \hat{\beta}_m = \text{Var}(S),$$

the total variance of the locus score, so the residual variance simplifies to $\sigma_X^2 - \text{Var}(S)$.

Estimating σ_X^2 . The variance of the exposure X is not directly available from GWAS summary statistics, but can be recovered from the marginal association of any single SNP j with X . From the OLS model $X = \beta_{u,j} g_j + \varepsilon_j$,

$$\text{SE}(\hat{\beta}_{u,j})^2 = \frac{\sigma_X^2}{N \sigma_j^2} (1 - R_j^2) \approx \frac{\sigma_X^2}{N \sigma_j^2},$$

giving

$$\sigma_X^2 \approx \sigma_j^2 (N \text{SE}(\hat{\beta}_{u,j})^2 + \hat{\beta}_{u,j}^2). \quad (9)$$

Taking the median of equation (9) over all P SNPs in the locus reduces the influence of SNPs with unusually large or small marginal associations.

The standard error. Assembling the pieces:

$$\boxed{\text{SE}(\hat{\alpha}) = \frac{1}{\sqrt{\mathcal{I}_\alpha}} = \sqrt{\frac{\sigma_X^2 - \widehat{\text{Var}}(S)}{N \widehat{\text{Var}}(Z)}}.} \quad (10)$$

The z -score $\hat{\alpha} / \text{SE}(\hat{\alpha})$ tests whether the locus has any net association with the exposure. Because both $\hat{\alpha}$ and $\text{SE}(\hat{\alpha})$ scale proportionally with $\|\hat{\beta}_m\|$ (as shown by the formula), the z -score is scale-invariant: it is the same whether $\hat{\alpha} = \|\hat{\beta}_m\|$ (per-allele) or any positive scalar multiple thereof.

7 Instrument-outcome coefficient $\hat{\gamma}$

7.1 Setup

Let Y denote the outcome (binary or continuous) and let $\hat{\gamma}_u \in \mathbb{R}^P$ be the vector of **per-allele univariate** SNP-outcome coefficients from an outcome GWAS of N_Y individuals:

$$\hat{\gamma}_{u,j} = \frac{\text{Cov}(g_j, Y)}{\text{Var}(g_j)} = \frac{\text{Cov}(g_j, Y)}{\sigma_j^2}. \quad (11)$$

For a **binary** outcome (case-control logistic regression), $\hat{\gamma}_{u,j}$ is the per-allele log-odds ratio. If only a z -score $z_j = \hat{\gamma}_{u,j} / \text{SE}(\hat{\gamma}_{u,j})$ is provided, the per-allele coefficient is recovered as

$$\hat{\gamma}_{u,j} = \frac{z_j}{\sqrt{N_Y \sigma_j^2 p(1-p)}}, \quad (12)$$

where p is the proportion of cases and $\sigma_j^2 = \text{Var}(g_j)$ is estimated from the reference panel.

7.2 Multivariate outcome coefficients from univariate summary statistics

By exactly the same argument as in Section 5, applying the inverse LD correlation matrix recovers the per-allele multivariate outcome coefficients:

$$\hat{\gamma}_m = \hat{\mathbf{D}}^{-1} \hat{\mathbf{C}}_g^{-1} \hat{\mathbf{D}} \hat{\gamma}_u. \quad (13)$$

These are the slopes that Y would receive in a joint regression of Y on all P SNPs simultaneously. Define the outcome score for individual i in the reference panel:

$$T_i = \mathbf{g}_i^\top \hat{\gamma}_m.$$

7.3 Reference-panel OLS estimator of $\hat{\gamma}$

With individual-level genotypes available in the reference panel but not in the outcome GWAS, we estimate $\hat{\gamma}$ by regressing the outcome score T on the scalar instrument Z through the origin:

$$\hat{\gamma} = \frac{\sum_{i=1}^{n_{\text{ref}}} T_i Z_i}{\sum_{i=1}^{n_{\text{ref}}} Z_i^2}. \quad (14)$$

Justification. In the population, $\mathbb{E}[Y | \mathbf{g}] \approx \mathbf{g}^\top \gamma_m = T$, so (14) approximates $\text{Cov}(Y, Z) / \text{Var}(Z)$, which is the instrument-outcome regression coefficient.

Closed form. Expanding $T_i = \mathbf{g}_i^\top \hat{\gamma}_m$ and $Z_i = \mathbf{g}_i^\top \hat{\beta}_m / \hat{\alpha}$:

$$\begin{aligned} \sum_i T_i Z_i &= \frac{1}{\hat{\alpha}} \sum_i (\mathbf{g}_i^\top \hat{\gamma}_m) (\mathbf{g}_i^\top \hat{\beta}_m) = \frac{n_{\text{ref}}}{\hat{\alpha}} \hat{\gamma}_m^\top \hat{\Sigma}_g \hat{\beta}_m, \\ \sum_i Z_i^2 &= \frac{1}{\hat{\alpha}^2} \sum_i (\mathbf{g}_i^\top \hat{\beta}_m)^2 = \frac{n_{\text{ref}}}{\hat{\alpha}^2} \hat{\beta}_m^\top \hat{\Sigma}_g \hat{\beta}_m = \frac{n_{\text{ref}}}{\hat{\alpha}^2} \widehat{\text{Var}}(S). \end{aligned}$$

Therefore

$$\hat{\gamma} = \hat{\alpha} \frac{\hat{\gamma}_m^\top \hat{\Sigma}_g \hat{\beta}_m}{\hat{\beta}_m^\top \hat{\Sigma}_g \hat{\beta}_m}. \quad (15)$$

Using $\hat{\gamma}_m = \hat{\mathbf{D}}^{-1} \hat{\mathbf{C}}_g^{-1} \hat{\mathbf{D}} \hat{\gamma}_u$ and $\hat{\Sigma}_g = \hat{\mathbf{D}} \hat{\mathbf{C}}_g \hat{\mathbf{D}}$:

$$\hat{\gamma}_m^\top \hat{\Sigma}_g \hat{\beta}_m = \hat{\gamma}_u^\top \hat{\mathbf{D}} \hat{\mathbf{C}}_g^{-1} \hat{\mathbf{C}}_g \hat{\mathbf{D}} \hat{\beta}_m = \hat{\gamma}_u^\top \hat{\mathbf{D}}^2 \hat{\beta}_m = \sum_{j=1}^P \hat{\sigma}_j^2 \hat{\gamma}_{u,j} \hat{\beta}_{m,j}.$$

Substituting and writing $\widehat{\text{Var}}(S) = \hat{\beta}_m^\top \hat{\Sigma}_g \hat{\beta}_m$:

$$\boxed{\hat{\gamma} = \hat{\alpha} \frac{\sum_{j=1}^P \hat{\sigma}_j^2 \hat{\gamma}_{u,j} \hat{\beta}_{m,j}}{\widehat{\text{Var}}(S)}}. \quad (16)$$

This shows that $\hat{\gamma}$ is a weighted sum of the per-allele products $\hat{\gamma}_{u,j} \hat{\beta}_{m,j}$, with weights $\hat{\sigma}_j^2$ proportional to the genotype variance of each SNP.

7.4 Standard error of $\hat{\gamma}$

For a **binary** outcome with proportion p of cases, the Fisher information for the slope γ in a logistic regression on Z is

$$\mathcal{I}_\gamma = N_Y \text{Var}(Z) p(1-p),$$

giving

$$\boxed{\text{SE}(\hat{\gamma}) = \frac{1}{\sqrt{\mathcal{I}_\gamma}} = \frac{1}{\sqrt{N_Y \widehat{\text{Var}}(Z) p(1-p)}} = \frac{\hat{\alpha}}{\sqrt{N_Y \widehat{\text{Var}}(S) p(1-p)}}.} \quad (17)$$

For a **continuous** outcome with residual variance σ_Y^2 , $\mathcal{I}_\gamma = N_Y \text{Var}(Z)/\sigma_Y^2$ and $\text{SE}(\hat{\gamma}) = \sigma_Y/\sqrt{N_Y \text{Var}(Z)}$.

Note that $\text{SE}(\hat{\gamma})$ depends on $\widehat{\text{Var}}(Z) = \widehat{\text{Var}}(S)/\hat{\alpha}^2$ computed from the reference panel, the outcome sample size N_Y , and (for binary outcomes) the case fraction p . It does not require knowledge of σ_Y^2 .

8 Handling near-singular LD matrices

When $P \gg n_{\text{ref}}$ or SNPs are in near-perfect LD, $\hat{\mathbf{C}}_g$ is ill-conditioned. The **truncated pseudoinverse** stabilises the computation.

Write the eigendecomposition

$$\hat{\mathbf{C}}_g = \mathbf{V} \text{diag}(d_1, \dots, d_P) \mathbf{V}^\top, \quad d_1 \geq \dots \geq d_P \geq 0,$$

and retain only the K eigenvalues with $d_k > \epsilon_0 d_1$ (default $\epsilon_0 = 0.01$). The pseudoinverse is

$$\hat{\mathbf{C}}_g^+ = \mathbf{V}_K \text{diag}(d_1^{-1}, \dots, d_K^{-1}) \mathbf{V}_K^\top.$$

The regularised estimate is then

$$\hat{\beta}_m = \hat{\mathbf{D}}^{-1} \hat{\mathbf{C}}_g^+ \hat{\mathbf{D}} \hat{\beta}_u. \quad (8)$$

The effective rank K measures the number of independent dimensions of LD variation in the locus.

9 Summary: the complete algorithm

Instrument-exposure coefficients $\hat{\alpha}$

1. **Input:** per-allele marginal exposure coefficients $\hat{\beta}_{u,j}$ and standard errors $\hat{s}_j$ from the exposure GWAS (N individuals); reference-panel SDs $\hat{\sigma}_j$ (or allele frequencies $\hat{f}_j$).
2. Form $\hat{\mathbf{D}} = \text{diag}(\hat{\sigma}_1, \dots, \hat{\sigma}_P)$; compute $\hat{\mathbf{C}}_g$ from the reference panel; invert via truncated eigendecomposition (threshold ϵ_0) to get $\hat{\mathbf{C}}_g^+$.
3. Apply eq. (8): $\hat{\beta}_m = \hat{\mathbf{D}}^{-1} \hat{\mathbf{C}}_g^+ \hat{\mathbf{D}} \hat{\beta}_u$.
4. Set $\hat{\alpha} = \|\hat{\beta}_m\|$ and $Z_i = \mathbf{g}_i^\top \hat{\beta}_m / \hat{\alpha}$.
5. Estimate $\widehat{\text{Var}}(S)$ from reference-panel scores $S_i = \mathbf{g}_i^\top \hat{\beta}_m$; estimate $\hat{\sigma}_X^2$ via eq. (9); compute $\text{SE}(\hat{\alpha})$ via eq. (10).

Instrument-outcome coefficients $\hat{\gamma}$

1. **Input:** per-allele marginal outcome coefficients $\hat{\gamma}_{u,j}$ from the outcome GWAS (N_Y individuals, case fraction p for binary outcomes). If only z -scores are available, recover $\hat{\gamma}_{u,j}$ via eq. (12).
2. Apply eq. (13) using the *same* $\hat{\mathbf{D}}$, $\hat{\mathbf{C}}_g^+$ as for the exposure: $\hat{\gamma}_m = \hat{\mathbf{D}}^{-1} \hat{\mathbf{C}}_g^+ \hat{\mathbf{D}} \hat{\gamma}_u$.
3. Compute outcome scores $T_i = \mathbf{g}_i^\top \hat{\gamma}_m$ in the reference panel.
4. Estimate $\hat{\gamma}$ via eq. (14) (reference-panel OLS of T on Z), or equivalently via the closed form eq. (16).
5. Compute $\text{SE}(\hat{\gamma})$ via eq. (17).

The central result, equation (7), states that premultiplying the vector of marginal per-allele coefficients by $\hat{\mathbf{D}}^{-1} \hat{\mathbf{C}}_g^{-1} \hat{\mathbf{D}}$ — equivalently, scaling to per-SD units, premultiplying by the inverse correlation matrix, and rescaling back — recovers the joint multivariate per-allele coefficients without requiring individual-level data.

Part II

GATE score coefficient from summary-level pQTL statistics

10 Background

The **GATE** (Genomewide Aggregated Trans- Effects) method constructs a composite genetic score for an exposure by summing the genetically predicted exposure values over all trans-pQTLs, then tests this GATE score for association with a disease outcome. This part derives a formula that computes the log-OR per SD of the GATE score—and its standard error—directly from the per-locus summary statistics ($\hat{\alpha}_k, \hat{\gamma}_k, \text{SE}(\hat{\gamma}_k), \text{sd}_{Z,k}$) that are already available from the MR-Hevo pipeline, without requiring access to individual-level data.

11 Notation

Symbol	Meaning
k	Trans-pQTL locus index; $k = 1, \dots, K$
j	SNP index within locus k
G_{ij}	Genotype (0/1/2 copies of effect allele) for individual i , SNP j
$\hat{\beta}_{m,kj}$	Per-allele instrument weight for SNP j in locus k (from UKBB zarr LD)
$Z_{ik} = \sum_j G_{ij} \hat{\beta}_{m,kj}$	Scalar instrument score for individual i at locus k
$\hat{\alpha}_k$	Instrument-exposure coefficient: $E[\Delta X \mid \Delta Z_k = 1]$
$\hat{\gamma}_k$	Instrument-outcome coefficient: $E[\Delta \log \text{OR}(Y) \mid \Delta Z_k = 1]$
$\text{SE}(\hat{\gamma}_k)$	Standard error of $\hat{\gamma}_k$ (from LD-approximate formula, eq. (17))
$\text{sd}_{Z,k}$	SD of Z_k in the UKBB reference panel (stored as <code>sd.Z</code>)
N, p	Total sample size and case fraction in the target dataset

12 The GATE score

The GATE score for individual i , summed over all trans-pQTLs, is

$$G_i = \sum_{k=1}^K \hat{\alpha}_k Z_{ik},$$

where the cis-pQTL is excluded. Trans-pQTL loci are in distinct genomic regions and are therefore approximately uncorrelated, so

$$\text{Var}_{\text{ref}}(G) = \sum_k \hat{\alpha}_k^2 \text{sd}_{Z,k}^2$$

with no cross-terms. The standardised GATE score is

$$G_i^* = G_i / \text{SD}_{\text{ref}}(G).$$

13 Score test for $\hat{\beta}_{\text{GATE}}$

13.1 Score and information

Let $Y_i \in \{0, 1\}$ denote disease status and let $\hat{\mu}_i^{(0)}$ denote the fitted probability under the null logistic model (intercept + covariates, no genetic term). Define

$$r_i = Y_i - \hat{\mu}_i^{(0)}, \quad w_i = \hat{\mu}_i^{(0)}(1 - \hat{\mu}_i^{(0)}).$$

The score and partial Fisher information for a scalar instrument X are

$$U_X = \mathbf{x}^\top \mathbf{r}, \quad V_X = \mathbf{x}^\top \mathbf{W} \mathbf{x} - \mathbf{x}^\top \mathbf{W} \mathbf{Z} (\mathbf{Z}^\top \mathbf{W} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{W} \mathbf{x},$$

where $\mathbf{Z}$ is the null-model design matrix (intercept + covariates) and $\mathbf{W} = \text{diag}(w_i)$. The score estimate is $\hat{\beta} = U_X / V_X$ with $\text{SE}(\hat{\beta}) = 1 / \sqrt{V_X}$.

Since the null model includes an intercept and all covariates, the per-locus score $U_k = \mathbf{z}_k^\top \mathbf{r}$ already adjusts for their correlations with Z_k .

13.2 Combining over trans-pQTLs

For the standardised composite $G^* = G / \text{SD}_G = (\sum_k \hat{\alpha}_k Z_k) / \text{SD}_G$:

$$U_{G^*} = \frac{1}{\text{SD}_G} \sum_k \hat{\alpha}_k U_k, \quad V_{G^*} = \frac{1}{\text{SD}_G^2} \sum_k \hat{\alpha}_k^2 V_k,$$

using the independence of instruments across loci ($U_k \perp U_l$, $V_k \perp V_l$ for $k \neq l$).

The score estimate is

$$\hat{\beta}_{\text{GATE}} = \frac{U_{G^*}}{V_{G^*}} = \text{SD}_G \cdot \frac{\sum_k \hat{\alpha}_k U_k}{\sum_k \hat{\alpha}_k^2 V_k}.$$

Substituting $U_k = \hat{\gamma}_k V_k$ (the score estimate for locus k):

$$\hat{\beta}_{\text{GATE}} = \text{SD}_G \cdot \frac{\sum_k \hat{\alpha}_k \hat{\gamma}_k / \text{SE}(\hat{\gamma}_k)^2}{\sum_k \hat{\alpha}_k^2 / \text{SE}(\hat{\gamma}_k)^2} \quad (18)$$

$$\text{SE}(\hat{\beta}_{\text{GATE}}) = \frac{\text{SD}_G}{\sqrt{\sum_k \hat{\alpha}_k^2 / \text{SE}(\hat{\gamma}_k)^2}} \quad (19)$$

14 Simplification of the standard error

The LD-approximate SE formula (eq. (17) from Part-I) gives

$$\text{SE}(\hat{\gamma}_k) = \frac{1}{\sqrt{N \text{sd}_{Z,k}^2 p(1-p)}}, \quad \text{so} \quad V_k = \frac{1}{\text{SE}(\hat{\gamma}_k)^2} = N \text{sd}_{Z,k}^2 p(1-p).$$

Substituting into the denominator of eq. (19):

$$\sum_k \hat{\alpha}_k^2 V_k = N p(1-p) \sum_k \hat{\alpha}_k^2 \text{sd}_{Z,k}^2 = N p(1-p) \text{SD}_G^2.$$

Therefore

$$\text{SE}(\hat{\beta}_{\text{GATE}}) = \frac{\text{SD}_G}{\sqrt{N p(1-p)} \text{SD}_G} = \frac{1}{\sqrt{N p(1-p)}}.$$

The SE is determined entirely by the sample size and case fraction, equal to the SE from a single 1-df binary logistic regression.

15 Equivalence to the IVW z-statistic

Substituting $V_k = N \text{sd}_{Z,k}^2 p(1-p)$ and defining the per-SD(Z) quantities $\hat{\alpha}_k^* = \hat{\alpha}_k \text{sd}_{Z,k}$ and $\hat{\gamma}_k^* = \hat{\gamma}_k \text{sd}_{Z,k}$ (the values displayed in the instrument tables), the z-statistic becomes

$$z_{\text{GATE}} = \frac{\hat{\beta}_{\text{GATE}}}{\text{SE}(\hat{\beta}_{\text{GATE}})} = \sqrt{N p(1-p)} \cdot \frac{\sum_k (\hat{\alpha}_k^*)^2 \hat{\theta}_k}{\sqrt{\sum_k (\hat{\alpha}_k^*)^2}},$$

where $\hat{\theta}_k = \hat{\gamma}_k^* / \hat{\alpha}_k^*$ is the per-locus IV-ratio estimate. This is the IVW z-statistic with weights $w_k = (\hat{\alpha}_k^*)^2$ proportional to the reciprocal variance of the ratio estimator—identical to the MR-Hevo IVW estimator restricted to trans-pQTLs.

16 Comparison: summary-level vs direct regression (PDCD1/T1D)

The summary-level formula is validated against the directly computed coefficient from logistic regression of T1D on the individual-level GATE score in the SDRNT1BIO/GS dataset (4,922 cases, 7,452 controls). For the direct computation, the GATE score is accumulated as $G_i = \sum_{k,j} G_{ij} \cdot (\hat{\alpha}_k \hat{\beta}_{m,kj})$ across all trans-pQTL SNPs in the individual-level genotype data on **diabepi**.

The two methods differ in the SD used for standardisation: the summary-level formula uses $\text{SD}_{\text{ref}}(G)$ from the UKBB reference panel, while the direct regression standardises by the empirical SD in the target GS dataset. The z-statistics are invariant to this scaling difference; the ratio $\hat{\beta}_{\text{summary}} / \hat{\beta}_{\text{direct}}$ estimates $\text{SD}_{\text{ref}}(G) / \text{SD}_{\text{target}}(G)$ and should be close to 1 for a well-calibrated reference panel.

Table 2: GATE coefficient for T1D ($\hat{\beta}_{\text{GATE}} = \text{log-OR per SD of GATE score}$) from summary-level formula (eqs. 18–19) vs direct individual-level regression. Trans-pQTLs: number of pQTL loci contributing to the GATE score (cis-pQTL PDCD1 excluded). SE (summary-level) = $1/\sqrt{N p(1-p)} \approx 0.0184$; SE (direct) uses the empirical SD of the GATE score in the GS target dataset.

Method	$\hat{\beta}$	SE	z	p	Trans pQTLs
Summary-level	0.2244	0.01837	12.218	2.5e-34	38
Direct (individual-level)	0.1965	0.02020	9.729	2.27e-22	38

The z-statistics from both methods agree within the range expected from LD mismatch between the UKBB reference panel and the GS target dataset, analogous to the attenuation seen in the LD-approximate vs direct $\hat{\gamma}$ comparison in the main report (Pearson $r \approx 0.83$).